Unit:-1
Introduction to Cloud Computing

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Pre-requisite : Fundamentals of Distributed Computing

- A distributed computer system consists of multiple software components that are on multiple computers, but run as a single system.
- The computers that are in a distributed system can be physically close together and connected by a local network, or they can be geographically distant and connected by a wide area network.
- A distributed system can consist of any number of possible configurations, such as mainframes, personal computers, workstations, minicomputers, and so on.
- The goal of distributed computing is to make such a network work as a single computer.



Cloud Computing

- Cloud Computing is defined as storing and accessing data and computing services over the Internet.
- It is the on-demand availability of computer services like servers, networking, databases, data storage, etc.
- The main goal of cloud computing is to give data centers access to many users for accessing data from remote servers.



Advantages

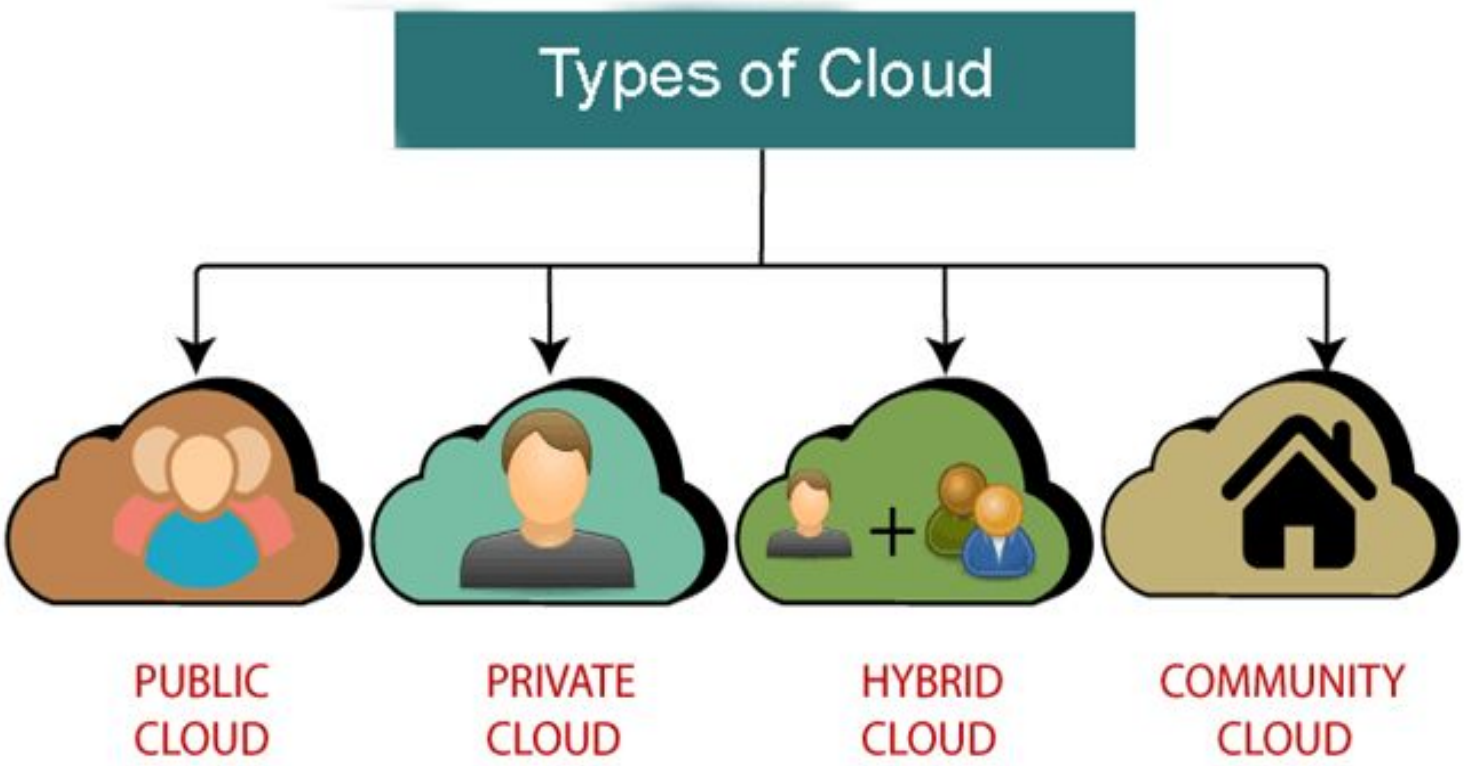
- Back-up and restore data
- Improved collaboration
- Excellent accessibility
- Low maintenance cost
- Mobility
- Unlimited storage capacity
- Data security



Disadvantages

- Internet Connectivity
- Vendor lock-in
- Limited Control
- Security




Types of Cloud
PUBLIC CLOUD
PRIVATE CLOUD
HYBRID CLOUD
COMMUNITY CLOUD

Public Cloud

- Public cloud is **open to all** to store and access information via the Internet using the pay-per-usage method.
- In public cloud, computing resources are managed and operated by the Cloud Service Provider (CSP).
- **Example:** Amazon elastic compute cloud (EC2), IBM SmartCloud Enterprise, Microsoft, Google App Engine, Windows Azure Services Platform



Advantages of Public Cloud

- Public cloud is owned at a lower cost than the private and hybrid cloud.
- Public cloud is maintained by the cloud service provider, so do not need to worry about the maintenance.
- Public cloud is easier to integrate. Hence it offers a better flexibility approach to consumers.
- Public cloud is location independent because its services are delivered through the internet.
- Public cloud is highly scalable as per the requirement of computing resources.
- It is accessible by the general public, so there is no limit to the number of users.



Disadvantages of Public Cloud

- Public Cloud is less secure because resources are shared publicly.
- Performance depends upon the high-speed internet network link to the cloud provider.
- The Client has no control of data.



Private Cloud

- Private cloud is also known as an **internal cloud** or **corporate cloud**.
- It is used by organizations to build and manage their own data centers internally or by the third party.
- It can be deployed using Opensource tools such as Openstack and Eucalyptus.



Advantages of Private Cloud

- Private cloud provides a high level of security and privacy to the users.
- Private cloud offers better performance with improved speed and space capacity.
- It allows the IT team to quickly allocate and deliver on-demand IT resources.
- The organization has full control over the cloud because it is managed by the organization itself. So, there is no need for the organization to depends on anybody.
- It is suitable for organizations that require a separate cloud for their personal use and data security is the first priority.



Disadvantages of Private Cloud

- Skilled people are required to manage and operate cloud services.
- Private cloud is accessible within the organization, so the area of operations is limited.
- Private cloud is not suitable for organizations that have a high user base, and organizations that do not have the prebuilt infrastructure, sufficient manpower to maintain and manage the cloud.



Hybrid Cloud

- Hybrid Cloud is a combination of the public cloud and the private cloud. we can say:
- ***Hybrid Cloud = Public Cloud + Private Cloud***
- Hybrid cloud is partially secure because the services which are running on the public cloud can be accessed by anyone, while the services which are running on a private cloud can be accessed only by the organization's users.
- **Example:** Google Application Suite (Gmail, Google Apps, and Google Drive), Office 365 (MS Office on the Web and One Drive), Amazon Web Services.



Advantages of Hybrid Cloud

- Hybrid cloud is suitable for organizations that require more security than the public cloud.
- Hybrid cloud helps you to deliver new products and services more quickly.
- Hybrid cloud provides an excellent way to reduce the risk.
- Hybrid cloud offers flexible resources because of the public cloud and secure resources because of the private cloud.



Disadvantages of Hybrid Cloud

- In Hybrid Cloud, security feature is not as good as the private cloud.
- Managing a hybrid cloud is complex because it is difficult to manage more than one type of deployment model.
- In the hybrid cloud, the reliability of the services depends on cloud service providers.



Community Cloud

- Community cloud allows systems and services to be accessible by a group of several organizations to share the information between the organization and a specific community.
- It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.
- **Example:** Health Care community cloud



Advantages of Community Cloud

- Community cloud is cost-effective because the whole cloud is being shared by several organizations or communities.
- Community cloud is suitable for organizations that want to have a collaborative cloud with more security features than the public cloud.
- It provides better security than the public cloud.
- It provides collaborative and distributive environment.
- Community cloud allows us to share cloud resources, infrastructure, and other capabilities among various organizations.



Disadvantages of Community Cloud

- Community cloud is not a good choice for every organization.
- Security features are not as good as the private cloud.
- It is not suitable if there is no collaboration.
- The fixed amount of data storage and bandwidth is shared among all community members.



Cloud Infrastructure

Cloud infrastructure consists of all of the hardware and software elements needed for cloud computing, including:

- Compute (server)
- Networking
- Storage
- Virtualization resources



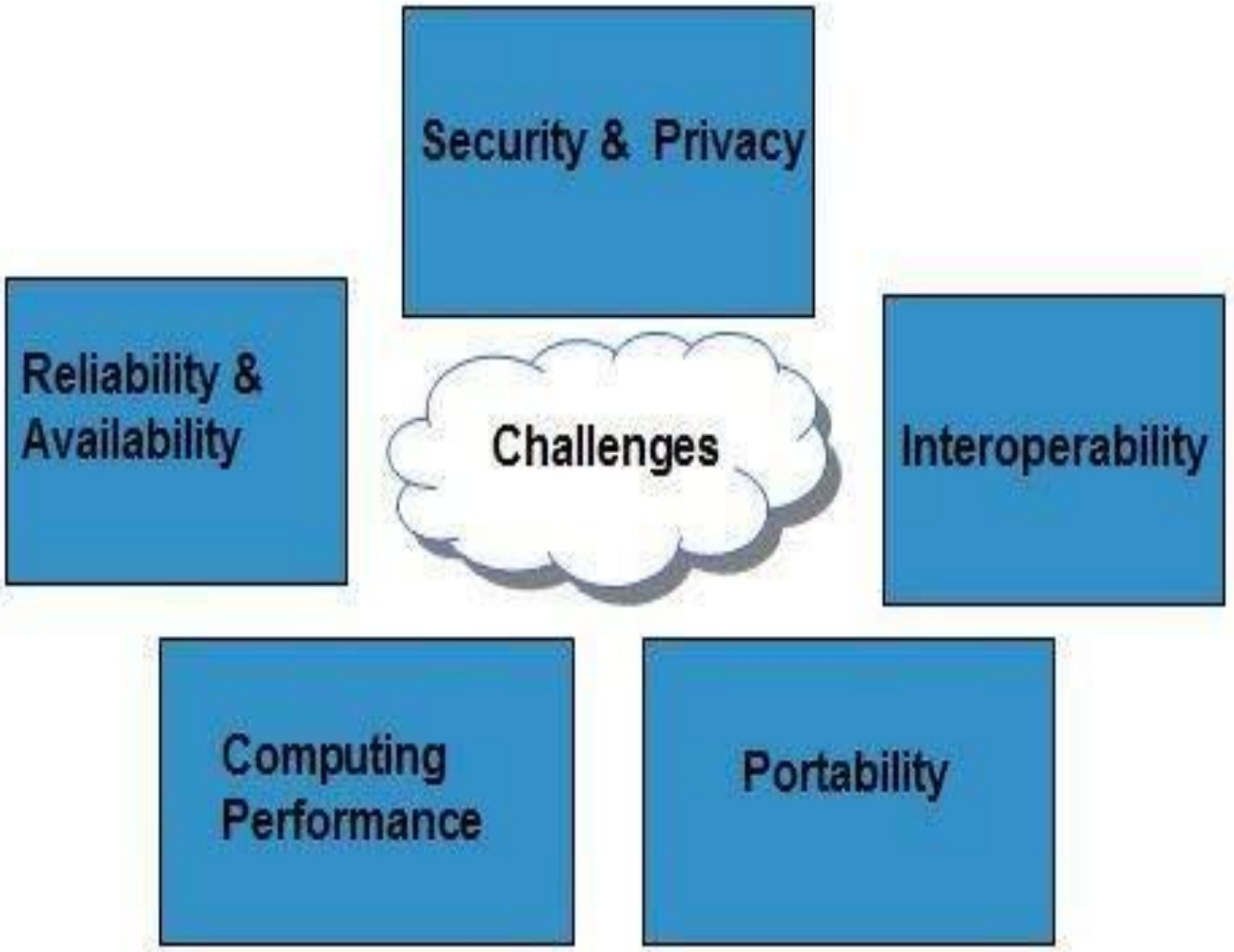
Cloud infrastructure management

- Cloud infrastructure management comprises the processes and tools needed to effectively allocate and deliver key resources when and where they are required.
- The UI, or dashboard, is a good example of such a tool; it acts as a control panel for provisioning, configuring and managing cloud infrastructure.

Cloud infrastructure management is useful in delivering cloud services to both:

- Internal users, such as developers or any other roles that consume cloud resources.
- External users, such as customers and business partners.




Security & Privacy
Reliability & Availability
Challenges
Interoperability
Computing Performance
Portability

Cloud Computing Application



Virtualization in Cloud Computing

- **Virtualization** is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- In other words, Virtualization is a technique, which allows to share a single physical instance of a resource or an application among multiple customers and organizations.
- It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.



Types of Virtualization

- Hardware Virtualization.
- Operating system Virtualization.
- Server Virtualization.
- Storage Virtualization.



Hardware Virtualization

- When the virtual machine software or virtual machine manager *(VMM) is directly installed on the hardware system* is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.
- After virtualization of hardware system we can install different operating system on it and run different applications on those OS.

Usage:

- Hardware virtualization is mainly done for the server platforms, because controlling virtual machines is much easier than controlling a physical server.



Operating System Virtualization

- When the virtual machine software or virtual machine manager *(VMM) is installed on the Host operating system* instead of directly on the hardware system is known as operating system virtualization.

Usage:

- Operating System Virtualization is mainly used for testing the applications on different platforms of OS.



Server Virtualization

- When the virtual machine software or virtual machine manager *(VMM) is directly installed on the Server system* is known as server virtualization.

Usage:

- Server virtualization is done because a single physical server can be divided into multiple servers on the demand basis and for balancing the load.



Storage Virtualization

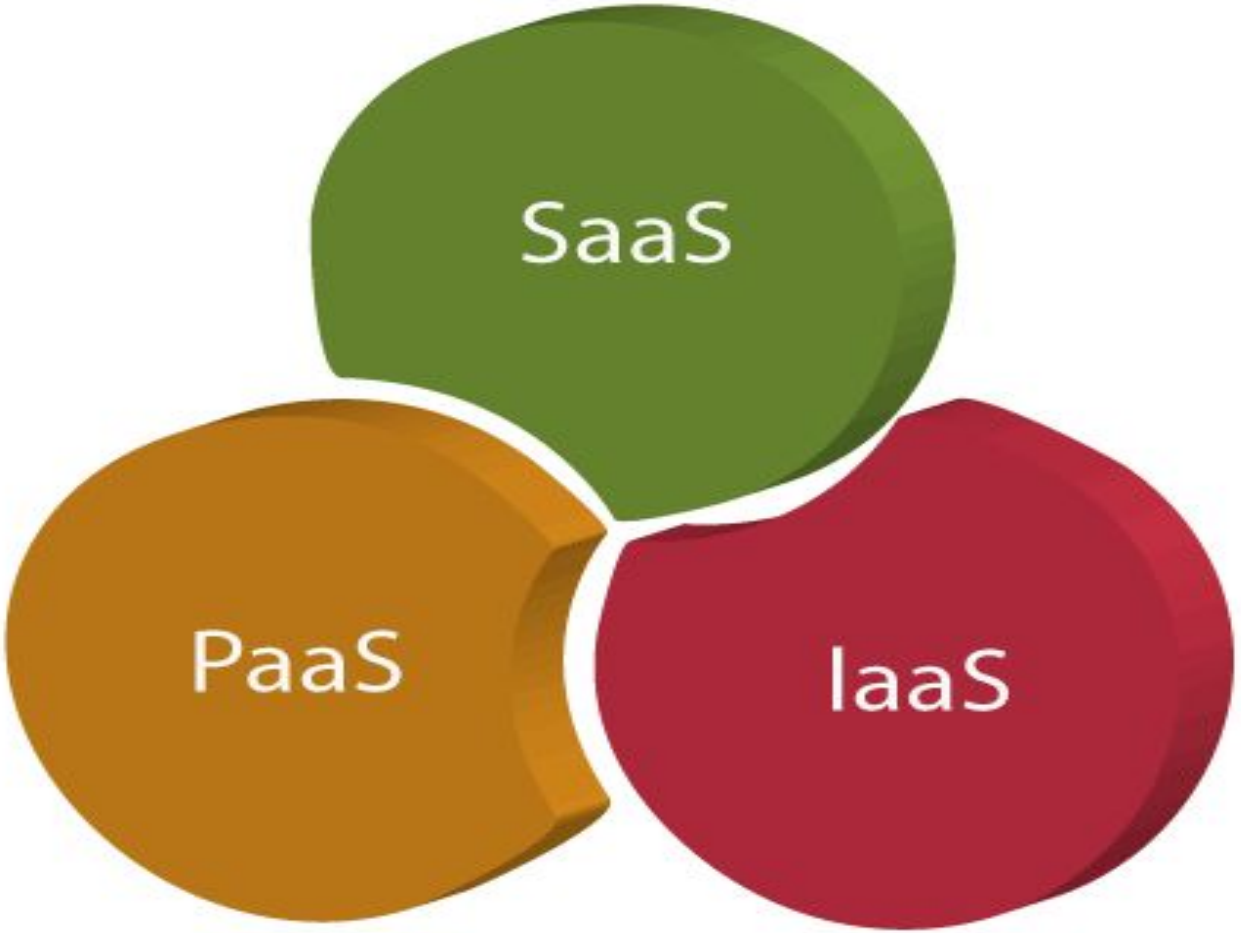
- Storage virtualization is the *process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device*.
- Storage virtualization is also implemented by using software applications.

Usage:

- Storage virtualization is mainly done for back-up and recovery purposes.



Cloud Service Models



Infrastructure as a Service (IaaS)

- IaaS is also known as **Hardware as a Service (HaaS)**.
- It is a computing infrastructure managed over the internet.
- The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS

- Resources are available as a service
- Services are highly scalable
- Dynamic and flexible
- GUI and API-based access
- Automated administrative tasks

Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.

Platform as a Service (PaaS)

- PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

  Characteristics of PaaS

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- Support multiple languages and frameworks.
- Provides an ability to "**Auto-scale**".

  Example: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.



Software as a Service (SaaS)

- SaaS is also known as "**on-demand software**".
- It is a software in which the applications are hosted by a cloud service provider.
- Users can access these applications with the help of internet connection and web browser.

Characteristics of SaaS

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis

Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

Thank You

